

TÜBTAK

TÜBTAK-2209-A UNIVERSITY STUDENTS RESEARCH PROJECTS SUPPORT PROGRAM

RESEARCH PROPOSAL FORM

[Yl] Year – [Dönem] Period Application

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A. GENERAL INFORMATION

Applicant's Full Name: [Örenci Ad Soyad]

Title of the Research Proposal: [Proje Bal]

Advisor's Full Name: [Danman Ad Soyad]

Institution/Organization Where the Research Will Be Conducted: [Üniversite/Kurum Ad]

ABSTRACT

[Projenin Türkçe veya İngilizce özeti buraya yazılacaktır. Konunun genel çerçevesi, projenin amacı, yöntemi ve beklenen sonuçlar kısaca özetlenmelidir...]

Keywords: [Anahtar Kelime 1], [Anahtar Kelime 2], [Anahtar Kelime 3]

ORIGINAL VALUE

Importance of the Subject, Original Value of the Research Proposal and Research Question/Hypothesis

1.1. Importance of the Subject and Scientific Excellence of the Project

AYBUStore addresses a structural access problem arising from the university's multi-campus organization. The service is currently physically available at only one location, while students are distributed across four campuses. This configuration creates unequal access to basic daily supplies, increases time and effort costs, and limits service continuity for students who are geographically distant from the store.

The project's scope is the digital transformation of this single-point physical service into a university-centered, hyper-local e-commerce system. Its boundary is intra-university use; it is designed specifically for the internal dynamics of the AYBU student ecosystem rather than for national-scale commerce. In this framework, the project is expected to increase service accessibility by at least 30% and to improve platform responsiveness through measurable Time to Interactive (TTI) optimization.

From a scientific perspective, the project targets a gap not sufficiently addressed by conventional e-commerce studies and platforms: the design of an equitable, closed-loop digital supply model for a specific multi-campus university context. Unlike general-market systems, AYBUStore proposes a context-specific architecture that integrates local demand patterns, campus distribution constraints, and continuous digital access requirements into a single service model.

The original contribution is fourfold. Conceptually, the project introduces a "Hyper-Local Closed-Loop Digital Ecosystem" for intra-university commerce. Methodologically, it replaces the traditional physical supply process with an inclusive and digitized service delivery model suitable for distributed campuses. Technically, it combines a modern front-end, scalable back-end, and embedded automated support modules in an end-to-end architecture. Practically, it enables equal, rapid, and location-independent access for students across all campuses on a 24/7 basis, thereby strengthening service equity and institutional integration.

1.2. Research Question and/or Hypothesis

Within a multi-campus university structure, a single-point physical market model produces unequal access, time loss, and operational inefficiency in meeting students' routine needs. This project investigates whether a hyper-local, closed-loop digital commerce model can eliminate these limitations.

The study addresses the following questions:

1. How can a university-specific hyper-local digital supply model be designed to provide continuous and equitable cross-campus access?
2. To what extent can a component-based, scalable architecture improve system responsiveness (TTI) relative to current static portal structures?

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3. How effectively can integrated support modules improve support speed and operational resolution performance?

The hypotheses are:

- **H1 (Inclusivity and Access):** AYBUStore will increase cross-campus service access by at least 30% and reduce spatial inequality compared with the current physical model.
- **H2 (Technical Performance):** The proposed modern architecture will improve Time to Interactive (TTI) by at least 40% versus the baseline static web portal and reduce average TTI below 1.5 seconds.
- **H3 (Operational Efficiency):** Integrated rule-based support modules (asynchronous chatbots, automated ticket routing, dynamic notifications) will reduce initial support response time by 60% and average issue resolution time by at least 45% relative to manual processes.

Hypotheses will be tested through a 3-month pilot after system completion. Log-based indicators (TTI, order process time, ticket response and resolution times) from at least 300 active students ($n \geq 300$), selected via stratified sampling, will be compared with historical baseline data from the existing system. Statistical significance will be evaluated using Independent Samples t-test (or Mann-Whitney U when distributional assumptions are not met), with significance threshold set at $p < 0.05$.

Aims and Objectives

1.3. Aim and Objectives of the Project

The main aim of this project is to develop AYBUStore as a hyper-local, closed-loop digital commerce ecosystem that removes physical access constraints in a multi-campus university setting and provides equitable, location-independent, and continuous service access.

To achieve this aim, the project defines the following measurable objectives for the 3-month pilot period:

- **Objective 1 (Access and Inclusivity):** Increase peripheral-campus utilization by at least 30% relative to the baseline (current estimated active access: 25% of total students) and reach at least 300 active verified student users across campuses.
- **Objective 2 (Technical Performance):** Reduce average Time to Interactive (TTI) from the baseline of approximately 3.5 seconds to below 1.5 seconds (minimum 40% improvement).
- **Objective 3 (Operational Efficiency):** Reduce first support response time from 12 hours by 60% and average issue-resolution time from 48 hours by at least 45% through integrated rule-based support modules.
- **Objective 4 (Reliability and Scalability):** Maintain 99.9% uptime and 0% transaction-level data loss during the pilot while sustaining peak load conditions of 500 requests/second and 1,000 concurrent active sessions.

Measurement is defined as follows: an active user is a unique SIS-verified student performing at least one in-system action within a 30-day window; utilization rate is calculated as

$$Utilization\ Rate\ (\%) = \left(\frac{U_{\text{active, peripheral}}}{U_{\text{total, peripheral}}} \right) \times 100$$

where $U_{\text{active, peripheral}}$ denotes active peripheral-campus users and $U_{\text{total, peripheral}}$ denotes total peripheral-campus students.

Reliability and performance will be monitored via continuous APM-based server logging (e.g., Prometheus/Grafana), and data integrity will be verified through PostgreSQL transaction/WAL logs and automated ACID compliance tests.

METHOD

This project follows a **mixed-method, design-and-evaluate** research design for a 3-month pilot: (i) iterative software development and deployment, (ii) log-based quantitative evaluation, and (iii) limited qualitative

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validation from user/support feedback. The method is selected because AYBUStore addresses both a *technical performance problem* and a *service equity problem*; therefore, only software implementation without statistical validation would be insufficient.

Methodology selection and rationale:

- **Agile/Scrum-based iterative development** is used to reduce delivery risk and support incremental validation of campus-specific requirements. This preference is aligned with higher-education digital transformation studies emphasizing process redesign and iterative institutional adaptation (Brdesee & Alsaggaf, 2022; Alenezi & Akour, 2023).
- **Performance-centered web engineering** (SSR-oriented optimization, measurable TTI tracking) is preferred because rendering strategy has a direct effect on interaction readiness and perceived quality (Netravali et al., 2018; Jahromi et al., 2020).
- **Rule-based support automation** is selected for operational predictability, fast first-level response, and suitability in education contexts (Güldal & Dinger, 2025; Dinh & Tran, 2023).
- **Hyper-local logistics design** is adopted to match campus-constrained distribution behavior and last-mile realities (Urquhart et al., 2022; Suryawanshi et al., 2021; Liu et al., 2021; Melkonyan et al., 2020).

Preliminary work (completed before pilot):

- Problem framing and scope definition for four-campus access inequality.
- Baseline extraction from current process assumptions and available historical indicators (access, TTI, support response/resolution times).
- Architecture design, KPI definition, and pilot instrumentation plan (TTI, order flow times, ticket metrics, uptime/data-integrity logs).

Research variables and measurement model:

- **Independent variable:** service model type (baseline physical/static model vs. AYBUStore digital model).
- **Dependent variables:** (a) cross-campus utilization rate, (b) Time to Interactive (TTI), (c) first support response time, (d) issue-resolution time, (e) uptime and transaction-level data loss.
- **Control factors:** pilot period (3 months), user group (SIS-verified students), and stratified campus sampling structure.

Sampling and data collection: At least $n \geq 300$ active students will be selected via stratified sampling across campuses. Data will be collected through system logs and monitoring tools (TTI telemetry, application/server logs, ticket lifecycle records, PostgreSQL transaction/WAL logs, and APM traces).

Statistical analysis:

- Descriptive statistics: mean, median, standard deviation, and percentage change from baseline.
- Hypothesis testing: Independent Samples t-test; Mann-Whitney U test when normality/homoscedasticity assumptions are not satisfied.
- Significance criterion: $p < 0.05$.

Method flow schema:

Preliminary diagnosis and baseline definition → Iterative development (Sprint cycles) → Pilot deployment on AYBU context → Log-based data collection → Statistical testing → Interpretation against H1–H3 and objective thresholds

This methodology is expected to provide both engineering validity (working system under campus constraints) and scientific validity (statistically testable evidence of improvement).

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PROJECT MANAGEMENT

Work-Time Schedule

WORK-TIME SCHEDULE

IP No	Name and Objectives of Work Packages	By whom(s) it will be carried out	Time Interval (Month)	Success Criteria and Contribution to the Success of the Project
1	[Paketi 1 Ad ve Amac]	[Sorumlu Kii(ler)]	Month [X-Y]	[Baar Kriteri 1]
2	[Paketi 2 Ad ve Amac]	[Sorumlu Kii(ler)]	Month [X-Y]	[Baar Kriteri 2]
3	[Paketi 3 Ad ve Amac]	[Sorumlu Kii(ler)]	Month [X-Y]	[Baar Kriteri 3]

Risk Management

RISK MANAGEMENT TABLE*

IP No	Most Significant Risks	Risk Management (Plan B)
1	[Olas Risk 1]	[B Plan / Çözüm Önerisi 1]
2	[Olas Risk 2]	[B Plan / Çözüm Önerisi 2]
3	[Olas Risk 3]	[B Plan / Çözüm Önerisi 3]

(*) The rows in the table can be expanded and reproduced as necessary.

3.3. Research Opportunities

TABLE OF RESEARCH OPPORTUNITIES (*)

Type and Model of Infrastructure/Equipment in the Organization (Laboratory, Vehicle, Machinery-Equipment, etc.)	Intended Use in the Project
[Altyap/Ekipman 1]	[Projedeki Kullanm Amac 1]
[Altyap/Ekipman 2]	[Projedeki Kullanm Amac 2]
[Altyap/Ekipman 3]	[Projedeki Kullanm Amac 3]

(*) The rows in the table can be expanded and reproduced as necessary.

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4. IMPACT

If implemented successfully, AYBUStore will convert the current single-location store model into a measurable campus-wide digital commerce ecosystem. The project is expected to generate scientific, technical, educational, and social outputs by improving cross-campus access equity, strengthening operational visibility for administrative units, and providing a reusable smart-campus software engineering case.

4.1. Expected Outputs

Output type	Expected Output(s)	Expected Time Interval
Scientific/Academic Outputs (National/International Article, Book, Book chapter, Proceeding etc.)	The project will produce an academic case study on a hyper-local closed-loop digital commerce ecosystem for multi-campus universities. Main outputs are the final technical report, architecture documentation, pilot evaluation results, and a student conference paper/article draft. Pilot data will provide measurable evidence on access improvement, TTI performance, and support-process efficiency.	0–12 months: final technical report, system design documentation and pilot evaluation outputs. 12–18 months: student conference paper or article draft.
Economical/Commercial/Social Outputs (Prototype, Product, Patent, Utility Model, Production License, Inventory/Database/Documentation Production, Spin-off/Start-up Company etc.)	The core output is the AYBUStore prototype/reference implementation including OTP authentication, catalogue, cart/order flow, simulated checkout, stock management, admin/CMS, campus-aware tracking, and rule-based support modules. Additional outputs are database/inventory schemas (users, products, categories, orders, support tickets, tracking), SRS/UML artefacts, test cases, and user/admin documentation. Socially, the project proposes a digital model that reduces access inequality caused by the current single-location structure.	0–12 months: prototype/reference implementation, database model, SRS, UML diagrams, test cases and user/admin documentation. Post-project: possible institutional adaptation by university units or extension into a broader smart-campus service model.
Outputs Regarding to Training Researcher and Creating New Project(s) (Graduate thesis, National/International new project)	The project will train undergraduate researchers in requirements engineering, Agile/Scrum planning, software architecture, cloud-based backend design, secure OTP, database modelling, testing, and performance evaluation. It also provides hypothesis-based research experience through pilot data, log-based measurement, and statistical comparison. These outputs can support follow-up projects on smart-campus services, AI-assisted support, campus-aware tracking, demand analysis, and higher-education digital transformation.	0–12 months: undergraduate researcher training and reusable project artefacts. 12–18 months / post-project: possible new student projects, graduation projects or institutional digital transformation initiatives.

(*) Time intervals are specified as 0–12 months, 12–18 months and post-project where applicable.

4.2. Expected Impacts

Expected Application Areas:

The primary application area of AYBUStore is **university-centered digital commerce and smart-campus service delivery**. It is designed for AYBU's distributed structure (Esenboa, Etlik, Bilkent, Çubuk) and can operate as a platform for campus-focused product discovery, order management, inventory visibility, support handling, and campus-aware tracking.

Potential end users are students (especially peripheral campuses), administrative units and store operators, IT units, and software engineering students/researchers. Students gain easier access to campus-related products; administrative units gain stronger product/stock/order/support visibility; IT units gain a reusable architecture reference; and researchers gain a practical case for requirements engineering, cloud architecture, secure authentication, usability, and performance evaluation.

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In the short term, outputs are expected to be used in AYBU as a prototype/reference implementation. In the medium/long term, the same model can be adapted by other multi-campus universities facing similar accessibility, inventory, and student-centered digital transformation challenges.

Socio-economic/Cultural Contribution:

AYBUStore is expected to contribute to **social equity and service accessibility** by reducing the access gap between campuses. In the current model, distance from the main store creates additional time, transportation, and opportunity costs. By digitizing discovery, ordering, tracking, and support processes, the project supports more equal participation in campus life.

From a socio-economic perspective, the project may reduce unnecessary travel for routine product access and improve efficiency through digital stock tracking, order monitoring, and administrative control. This supports data-driven decisions on product availability, campus demand, and service continuity.

From an educational perspective, the project contributes to training qualified undergraduate researchers in project management, requirements analysis, system design, secure authentication, cloud services, testing, risk management, and technical documentation.

The project also aligns with broader digital transformation and smart-campus goals by supporting student-centered service design, improved education-related service quality, inclusive institutional access, and scalable digital infrastructure in higher education.

4.3. Dissemination of Project Results and Activity Plan to be Implemented within the Scope of Science Communication

Field	Content
Target Group	Target groups are AYBU students, academic/administrative staff, IT units, campus store administrators, software engineering students, researchers, and other universities interested in smart-campus service models. Students benefit from improved access; institutional units benefit from architecture, database, documentation, and operational workflow outputs; researchers benefit from a real-world case study.
Goals and Expected Gains	The main goal is to increase awareness of how campus-specific digital commerce improves cross-campus accessibility, operational efficiency, and student experience. Expected gains are stronger understanding of multi-campus service inequality, increased institutional awareness of smart-campus solutions, and reusable technical knowledge for future student projects and institutional digital transformation. Dissemination also supports stakeholder feedback collection.
Tools to be Used	Outputs will be shared through final presentation, technical report, prototype demonstration, UML and architecture documentation, screenshots, short demo video, and project poster/infographic. Subject to institutional approval, selected outputs may also be shared via department seminars, student project exhibitions, university digital channels, or a controlled GitHub/code archive.
Timing	Dissemination starts within 0–12 months during final presentation and project closure. Technical report, prototype demo, documentation, and evaluation outputs are shared at development completion. Additional stakeholder demonstrations may follow prototype completion. Academic dissemination (conference paper/article draft) is targeted for 12–18 months .

ANY OTHER ISSUES YOU WOULD LIKE TO MENTION

[Buraya projeye ilgili eklemek istediiniz dier detaylar, etik kurallar, izin belgeleri veya destek alnan farkl hususlar eklenecektir...]

APPENDICES ANNEX-1: RESOURCES

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